November 26, 2025

Activity

1. Construct compare-ode driver and (also use euler, picard, predictor-corrector subroutines) to show solutions to the ODE equation":

   a)
   $$\frac{d^2y}{dx^2} + \beta\frac{dy}{dx} + \omega_0^2 y = 0,$$

   b)
   $$\frac{d^2y}{dx^2} + \beta\frac{dy}{dx} + \omega_0^2 y = \eta(t),$$

   where $\beta = 0.5$; $y(0) = 0.5$; $\dot{y}(0) = 1.0$; $\omega_0 = 1.0$; and η random noise in r∈[-1,1].

2. Construct compare-roots driver and (use newton & bisection subroutines) to show zeros of the function
   $$f(\mathrm{x}) = -\mathrm{x}^2 - 2\ln(\mathrm{x}) - 3\mathrm{e}^{-\mathrm{x}}$$
   Within an ending tolerance $\varepsilon = 2.0\times10^{-5}$,in the range [a,b]=[0.2,2.2].

3. Construct compare-differentiate driver and (also use FD, BD, MD subroutines) to show solutions to differentiation of the function
   $$f(\mathrm{x}) = \mathrm{x}^3 + 2\mathrm{x}^2 + 2\mathrm{x} + 2$$
   in the range [-5, 5].